

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2103

Course Title: Data Structures Lab

Assignment No: 01

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Lab Section: B1

Academic Semester: Spring 2025

Program: BSc in CSE

Question: Implement BFS and DFS

Solution:

Implement BFS

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
vector<int>V[100];
```

```
bool visited[100];
```

```
void BFS(int s)
```

```
{
```

```
    queue<int>q;
```

```
    visited[s] = true;
```

```
    q.push(s);
```

```
    while(!q.empty())
```

```
    {
```

```
        int node = q.front();
```

```
        q.pop();
```

```
        cout << node << " ";
```

```
        for(int i=0; i<V[node].size(); i++)
```

```
        {
```

```
            int next = V[node][i];
```

```
            if(!visited[next])
```

```
    {  
        visited[next] = true;  
        q.push(next);  
    }  
}  
}
```

```
int main()  
{  
    int n,e;  
    cout << "Number of nodes & edges : " ;  
    cin >> n >> e;  
  
    for(int i=0; i<e ; i++)  
    {  
        int nodeA, nodeB;  
        cin >> nodeA >> nodeB;  
  
        V[nodeA].push_back(nodeB);  
        V[nodeB].push_back(nodeA);  
    }
```

```
int s;  
cout << "Starting node : ";  
cin >> s;  
  
cout << "BFS Traversal : " ;  
BFS(s);  
  
return 0;  
}
```

INPUT:

Nodes: 7

Edges: 8

Edges list:

1 2

1 3

2 4

2 5

3 6

3 7

4 5

6 7

Starting node: 1

OUTPUT:

BFS Traversal : 1->2->3->4->5->6->7

Implement DFS

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
vector<int>V[100];
```

```
bool visited[100];
```

```
void DFS(int node)
```

```
{
```

```
    visited[node] = true;
```

```
    cout << node << " ";
```

```
    for(int i= 0; i< V[node].size(); i++)
```

```
    {
```

```
        int next = V[node][i];
```

```
        if(!visited[next])
```

```
        {
```

```
            DFS(next);
```

```
        }
```

```

    }
}

int main()
{
    int n,e;
    cout << "Number of nodes & edges : " ;
    cin >> n >> e;

    for(int i=0; i<e ; i++)
    {
        int nodeA, nodeB;
        cin >> nodeA >> nodeB;

        V[nodeA].push_back(nodeB);
        V[nodeB].push_back(nodeA);
    }

    int s;
    cout << "Starting node : ";
    cin >> s;

    cout << "DFS Traversal : " ;
    DFS(s);
}

```

```
    return 0;  
}
```

INPUT:

Nodes: 7

Edges: 8

Edges list:

1 2

1 3

2 4

2 5

3 6

3 7

4 5

6 7

Starting node: 1

OUTPUT:

DFS Traversal : 1->2->4->5->3->6->7